



This article explores Backbone.js with details about its Model, View and Router components. The author presents a few sample models to enlighten the reader.

Backbone.js is a JavaScript library, which helps you organise your client-side JavaScript code. It is based on the Model-View-Presenter architectural design pattern. It helps you to layer your JavaScript code and assign responsibility to each layer.

One of the reasons for the popularity of Backbone.js is its capability to communicate with the server through the REST API and keep the client-side data in sync with the server-side data, and vice versa. The modern Web is no longer browser-centric, but has to cater to a variety of clients like mobile devices, tablets and the apps within them. In this scenario, the REST API is the best option available to exchange data or content between client and server. Backbone.js's communication mechanism is built around the REST API.

Backbone.js consists of three major components:

1. Model (Collection)
2. View
3. Router

Each of these components, respectively, plays the role of a component of the Model-View-Presenter design.

Model

Model is used to represent the domain objects, which consist of data and logic to access, validate and convert the data. A sample model can be created as follows:

```
/* Declaring the Model for Book*/
```

```
var Book = Backbone.Model.extend({
  {
    urlRoot : '/books'}

  /*
    Initialize is the method which is called by default
    whenever an instance of the model created
    */
  initialize: function() {
    console.log("Book model has been created.");
  }
});

/*Instantiating an object, myBook. */
var myBook = new Book;
```

The *Initialize* method is called each time you create a new instance of the model. It is like a constructor, which you don't need to call explicitly but Backbone.js calls for you, while creating the object. The 'urlRoot' attribute comes handy when interacting with the server. The REST API calls are triggered on the path specified in the *urlRoot*, which is elaborated while communicating with the server as we go along.

You can set the attributes of the model as follows:

```
/* setting the attributes for the Model*/
var myBook = new Book;
```